10 (a) The Solar System includes the Sun and planets.

State **two** other types of natural object that orbit the Sun.

1 .....

2 .....

[2]

(b) State the shape of the orbits of the planets.

..... [1]

(c) Fig. 10.1 shows the orbit of an object around the Sun. At point A, the object is closest to the Sun. At point B, the object is furthest away from the Sun.

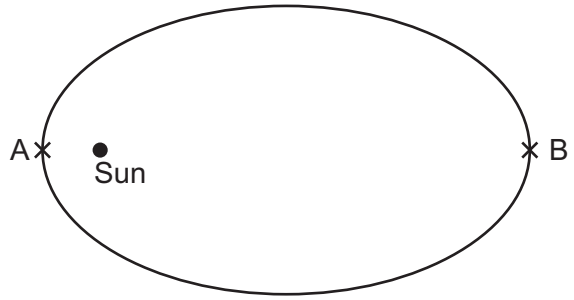

Fig. 10.1

State and explain the energy transfer as the object travels from point A to point B.

statement .....

.....

explanation .....

.....

[2]

(d) Jupiter is 7.8×10^{11} m from the Sun. The speed of light in a vacuum is 3.0×10^{8} m/s.

Calculate the time taken for light from the Sun to reach Jupiter.

time = ... [2]

[Total: 7]

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